

Transcutaneous electrical acupoint stimulation accelerates postoperative gastrointestinal function recovery in patients undergoing non-gastrointestinal laparoscopic surgery: a randomized controlled trial

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ABSTRACT

Background: and Purpose: Postoperative gastrointestinal dysfunction (POGD) is a significant complication following abdominal surgery and is characterized by delayed flatus and stool passage, postponed oral feeding, and dyspeptic symptoms. The study aimed to evaluate whether transcutaneous electrical acupoint stimulation (TEAS) accelerates gastrointestinal function recovery and reduces POGD-related complications after elective laparoscopic non-gastrointestinal surgery.

Materials and methods: Patients scheduled for selective non-gastrointestinal laparoscopic surgery were randomized into the TEAS or sham-TEAS group. TEAS group patients were administered electrical stimulation at acupoints Zusanli (ST36), Neiguan (PC6), and Hegu (LI4) starting 30 min prior to anesthesia induction and continuing until surgery completion. Patients in the sham-TEAS group received a sham stimulus at the same acupoints, and no output current was delivered by disconnecting the device's output line. Our primary endpoint included the time to first flatus, whereas the secondary endpoints were the time to first defecation, postoperative nausea and vomiting (PONV) rate, length of hospital stay, hospital expenses, numeric rating scale (NRS) pain score, and incidence of postoperative complications.

Results: A total of 587 patients were enrolled and assigned to the TEAS (n = 294) or sham-TEAS (n = 293) group between May 15, 2019, and October 18, 2021. Compared with the sham-TEAS group, the TEAS group had a significantly shorter time to first flatus (median [interquartile range (IQR)], 21.50 h [19.0–23.5 h] vs. 23.5 h [21.0–26.0 h]; $p < 0.001$) and time to first defecation (41.2 h [38.5–45.0 h] vs. 46.0 h [42.0–52.0 h]; $p < 0.001$). Additionally, the TEAS group showed lower rates of PONV (15% vs. 22.9%; $p = 0.015$) and abdominal distention (21.1% vs. 29.7%; $p = 0.017$), as well as lower NRS pain scores (median [IQR], 3 [2–3] vs. 3 [2–3.5]; $p = 0.026$). There were no significant differences in hospital stay length and hospitalization expenses.

Conclusion: TEAS accelerates gastrointestinal function recovery in patients undergoing laparoscopic non-gastrointestinal surgery, reducing PONV, abdominal distention and postoperative pain without increasing hospital stay or costs.

1. Introduction

Compared with open surgery, laparoscopic surgery offers advantages such as less surgical pain, shorter hospital stays, fewer infections, and faster return to normal activities [1–3]. However, the creation of pneumoperitoneum, the application of anaesthetics, and mechanical

punctures during laparoscopic procedures can disrupt gastrointestinal peristalsis. This disruption can subsequently lead to postoperative complications, including nausea, vomiting, abdominal distension, anal sphincter dysfunction, defecation difficulties, and delayed physical recovery [4,5]. Postoperative gastrointestinal dysfunction (POGD) significantly impairs patient recovery, prolongs hospital stays, and increases

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the risk of additional complications, with an estimated annual economic burden of approximately \$1.5 billion in the USA [6]. Postoperative ileus (POI), as the most common clinical manifestation of POGD, remains a critical concern for abdominal surgeons due to its high incidence and clinical impact [7].

The pathophysiology of POGD is complex and multifactorial, involving neurogenic, inflammatory, and pharmacological mechanisms, including sympathetic overactivation, cytokine release, and opioid use, all of which delay gastrointestinal motility [6]. It is generally believed that the early phase of the reaction to surgery is mediated neurally and involves neural reflexes activated during and immediately after surgery. An incision in the skin elevates adrenergic motor neuronal activity, leading to acute intestinal paralysis. After 3 to 4 h of surgical manipulation, the release of pro-inflammatory cytokines and chemokines up-regulates endothelial adhesion molecules. This cascade activates phagocytes residing throughout the gut, leading to leukocyte migration into the muscularis externa. Ultimately, these phagocytes release nitric oxide and prostaglandins, which directly inhibit smooth muscle contractility to prevent intestinal peristalsis [8]. In addition, the use of opioids, fluid overload, and gastrointestinal tract edema can all adversely affect gastrointestinal function, thereby prolonging recovery [7].

Therefore, current management strategies employ a multimodal approach that integrates pharmacological treatments with non-pharmacological interventions [8,9]. However, the high cost and potential adverse effects of pharmacological treatments can reduce their acceptability [10]. Alternative strategies focus on enhancing postoperative rehabilitation through interventions such as patient education, chewing gum, limited preoperative fasting, minimally invasive surgical techniques, early oral feeding, and effective analgesia [11–13]. Despite these approaches, no clinical guidelines or recommendations currently specify drugs or interventions tailored to preventing POGD.

In line with traditional Chinese medicine principles, surgical interventions are believed to induce Qi stagnation and blood stasis syndrome, disrupting the body's balance and leading to bowel dysfunction [14]. Acupuncture, through stimulation of specific acupoints, is proposed to regulate energy (Qi) flow and alleviate blood stasis, making it widely utilized for managing gastrointestinal disorders because of its efficacy and minimal invasion [14–16]. However, the reliance on skilled practitioners for traditional acupuncture and moxibustion limits its widespread application. Transcutaneous electrical acupoint stimulation (TEAS) offers a non-invasive alternative to conventional acupuncture, employing electrical stimulation through surface electrodes instead of needle insertion and manual manipulation, as in electroacupuncture. TEAS has demonstrated clinical benefits, including reduced intraoperative anesthetic requirements, decreased postoperative nausea and vomiting [5], alleviated postoperative pain [17], and improved postoperative recovery [18]. Moreover, previous research has shown that TEAS enhances gastrointestinal recovery by activating the vagal nerve and inhibiting inflammation, as evidenced by increased plasma levels of acetylcholine (ACh) and interleukin-10 (IL-10) alongside decreased levels of interleukin-6 (IL-6) and inducible nitric oxide synthase (iNOS) [19].

Although several rigorously designed randomized controlled trials (RCTs) [4,5,14,17,19,20] have evaluated TEAS for postoperative gastrointestinal dysfunction, there remains significant variability in the definitions of POI and POGD, as well as in clinical outcome measures for assessing gastrointestinal recovery. Furthermore, existing guidelines for the prevention and management of POGD have not yet incorporated recommendations for TEAS or electroacupuncture. Therefore, this study was conducted to evaluate whether TEAS, as an adjunctive non-pharmacological intervention, could effectively promote gastrointestinal function recovery and reduce the incidence of postoperative nausea and vomiting (PONV) and abdominal distension in patients undergoing elective laparoscopic non-gastrointestinal surgery.

2. Materials and Methods

2.1. Study design

The present prospective, double-blind, randomized controlled clinical trial was carried out at the tertiary Nankai Hospital from May 15, 2019, to October 18, 2021. The study was carried out with a TEAS group and a sham-TEAS group.

2.2. Patients

Eligible patients were those aged 18–64 years, with American Society of Anesthesiologists (ASA) class I–III disease, and planned to undergo selective laparoscopic non-gastrointestinal surgery. Patients with a previous history of abdominal surgery, gastrointestinal motility disorders, severe cardiovascular, hepatic, or renal disease, presence of a cardiac pacemaker, infection at acupuncture sites, allergy to acupuncture electrode patches, prior acupuncture treatment, or participation in other studies within three months preceding this trial were excluded.

2.3. Randomization and blinding

The principal investigator explained more details of this study to the enrolled patients one day prior to their scheduled surgery, in accordance with a standardized perioperative protocol. Laparoscopic non-gastrointestinal surgeries were conducted under general anesthesia. Patients who met the eligibility criteria were enrolled in this study and followed up postoperatively.

Patients were randomized into the TEAS or sham-TEAS groups at a 1:1 ratio using block randomization. The randomization sequence was generated by an independent statistician using computer software. The acupuncturist, the sole individual aware of treatment allocations, accessed the randomization sequence through sealed opaque envelopes containing computer-generated allocations. Patients, outcome assessors, and statisticians were blinded to group assignments.

2.4. Interventions

Acupoints were selected through a comprehensively review of the literature and expert consensus. Precise acupoint locations were determined using The National Standards for Acupoint Location (GB/T 12346–2021 nomenclature and location of meridian points) guidelines [21]. A licensed acupuncturist with nine years of clinical experience, registered as a traditional Chinese medicine (TCM) practitioner, independently administered either TEAS or sham-TEAS. The acupuncturist did not disclose group allocation but informed patients that tingling sensations might have occurred near the wrists and calves during treatment. In the TEAS group, acupoints associated with managing abdominal distension, abdominal pain, and constipation, such as Zusanli (stomach meridian ST36), Neiguan (pericardium meridian PC6), and Hegu (large intestine meridian LI4), were identified. The bilateral acupoint skin was cleaned with alcohol and dried, and electrodes were attached to the TEAS machine (Hua Tuo SDZ-III, Suzhou Medical Apparatus and Instruments Factory, Suzhou, China). The TEAS group was administered at a 2/10 Hz dense-disperse frequency, starting 30 min before anesthesia induction and continuing through the operation. We adjusted the stimulation intensity to induce mild muscle twitching at the individual's maximum tolerance, achieving the radiating paresthesia sensation known as *de qi* [22]. In the sham-TEAS group, identical acupoints were targeted, but stimulation was simulated by cutting output wires, ensuring no electrical current delivery [23] (eFigure in Supplement 1).

Standardized anesthesia protocols were employed across all patients. Patients received 5 mg of dexamethasone prior to induction, followed by 0.03 mg/kg midazolam, 1.5–2.5 mg/kg propofol, 0.3–0.4 µg/kg sufentanil, and 0.8 mg/kg rocuronium. Maintenance anesthesia consisted of

0.05–0.2 µg/kg/min remifentanyl and 4–12 mg/kg/h propofol, with intermittent rocuronium administration (0.2 mg/kg) every 30 min. Sedation depth was monitored using the bispectral index (BIS), which is a quantitative, dimensionless measure derived from processed electroencephalogram (EEG) signals and ranges from 0 to 100. BIS is routinely used to monitor the depth of anesthesia and sedation in real time, with values between 40 and 60 representing a state of general anesthesia [24]. In our study, the propofol infusion rate was adjusted to maintain BIS values between 40 and 60. The ventilation parameters were adjusted to maintain the pressure of end tidal carbon dioxide (PetCO₂) level at 30–40 mmHg. After peritoneal suturing, sufentanil (5–10 µg) was administered as needed. Upon skin closure, patients were transferred to the post-anesthesia care unit (PACU), where extubation was performed after spontaneous breathing resumed, with muscle relaxant antagonists (1–2 mg neostigmine and 0.5–1 mg atropine) administered as necessary. PONV were prevented by administering ondansetron, a 5-hydroxytryptamine-3 (5-HT₃) receptor antagonist. Postoperative analgesia included sufentanil and flurbiprofen via patient-controlled intravenous analgesia (PCIA) for 24–48 h if needed. Adverse events that occurred throughout this study were reported.

2.5. Outcome measures

Our primary endpoint included the time to first flatus, which referred to the duration between the completion of laparoscopic surgery and its occurrence (unit: hour); this was assessed every 6 h postoperatively until occurrence or discharge. The secondary endpoints were the time to first defecation (assessed at the same time points as the primary endpoint), pain score evaluated by the numeric rating scale (NRS; 0 = no pain, 10 = worst imaginable pain) at 6 h, 24 h, and 48 h postoperatively, and PONV incidence at the same three time points. Hospital expenses were documented in Chinese yuan (CNY) and converted into US dollars using an average exchange rate of 0.1525 from 2019 to 2021. Adverse events, whether related to the treatment or not, were reported through patient interviews conducted by a research assistant at each follow-up visit.

2.6. Statistical analysis

We determined our sample size in line with one preliminary work conducted in 2018 ($n = 50$), which showed times to first flatus of 21.5 ± 8.6 h and 23.5 ± 6.7 h for the TEAS and sham groups, respectively. Using a one-sided significance level of $\alpha = 0.025$ (corresponding to a two-sided $\alpha = 0.05$ for the final analysis) and 80% power, the minimum required sample size was determined to be 235 patients per group. To account for an anticipated dropout rate of about 20%, a total of 294 patients per group were planned. Sample size calculations were performed using PASS software, version 15.0 (NCSS, LLC, Kaysville, UT, USA).

Data analysis followed the per-protocol set principle. For the primary outcome, an intention-to-treat analysis was also performed. For missing data in the TEAS group, values were replaced with the maximum observed value (worst-case scenario) from the entire cohort. Conversely, missing data in the sham-TEAS group were replaced with the minimum observed value (best-case scenario) from the entire cohort. The normality of continuous variables was assessed using the Kolmogorov-Smirnov test. Continuous variables conforming to the normal distribution were presented as means with standard deviations (SDs) and were analyzed using two-tailed Student's *t*-tests. Continuous variables conforming to a non-normal distribution were presented as medians with interquartile ranges (IQRs) and were compared using Mann-Whitney *U* tests. Categorical variables were expressed as frequencies and proportions and analyzed using χ^2 tests or Fisher's exact tests. The cumulative no-flatus rate was assessed by log-rank tests and Kaplan-Meier curves. Comparison of the outcomes between the two groups was presented with effect sizes and their 95% confidence intervals (CIs) and *p*-values. A two-sided *p*-value less than 0.05 was considered statistically

significant. Statistical analysis was conducted using IBM SPSS 25.0 (IBM Corp., Armonk, NY, USA).

2.7. Ethical considerations

Ethics approval for this study was obtained from the Institutional Review Board of Tianjin Nankai Hospital (NKYY-YX-IRB-2017-035-02), and the study was conducted in accordance with the Declaration of Helsinki. This study was registered at [ClinicalTrials.gov](https://www.clinicaltrials.gov) (registration number NCT03724656). All patients provided written informed consent prior to enrollment and were informed of the possibility of withdrawing from the study at any time. Data were anonymized to protect privacy.

3. Results

3.1. Characteristics of patients

There were 1226 patients screened between May 2019 and October 2021 for eligibility. Of these, 612 were eliminated because of failure to meet the inclusion criteria, refusal to take part in this study, or other reasons. Ultimately, 614 eligible patients were allocated to the TEAS ($n = 308$) or sham-TEAS ($n = 306$) group. During this study, 12 and 9 patients in the TEAS and sham-TEAS groups, respectively, withdrew. Thus, 587 patients (TEAS: $n = 294$; sham-TEAS: $n = 293$) were included in this study and were thereby enrolled in the final analyses. The patient selection process is shown in Fig. 1. Patient demographic characteristics, comorbidities, surgical data and perioperative anesthetic use were compared between the two groups. The results showed that the two groups were well balanced on baseline and intra-operative data (Tables 1 and 2).

3.2. Primary endpoint

Patients in TEAS group demonstrated a remarkably shorter median time to first flatus compared to sham-TEAS group (21.5 h [19.0–23.5] vs. 23.5 h [21.0–26.0]; median difference: -2.5 h (95% CI, -3.0 to -1.8); $p < 0.001$). The intention-to-treat analysis found similar results: the time to first flatus in TEAS group compared to sham-TEAS group (21.3 h [19.0–23.5] vs. 23.4 h [21.0–26.0]; median difference: -2.0 h (95% CI, -2.9 to -1.5); $p < 0.001$) (Table 3). The cumulative no-flatus rate was assessed by log-rank tests and Kaplan-Meier curves. Kaplan-Meier analysis demonstrated a significantly higher cumulative no-flatus rate in the sham-TEAS group compared to the TEAS group (log-rank test, $P < 0.001$). The hazard ratio (HR) for the first flatus was 0.55 (95% CI, 0.45–0.67), indicating that patients receiving TEAS had a 45% higher probability of achieving first flatus at any given time point than those receiving sham stimulation (Fig. 2).

3.3. Secondary endpoints and adverse events

Compared with the sham-TEAS group, the TEAS group presented a significantly shorter median time to first defecation (41.2 h [38.5–45.0] vs. 46.0 h [42.0–52.0], $p < 0.001$). Additionally, the TEAS group demonstrated reduced 24 h PONV incidence (15.0% vs. 22.9%, $p = 0.015$), lower rates of abdominal distension (21.1% vs. 29.7%, $p = 0.017$), and decreased pain intensity (NRS pain score, 3 [2,3] vs. 3 [2–3.5], $p = 0.026$). The duration of postoperative hospital stay and hospitalization expenses were similar in both groups. Adverse events reported during the study were mild and resolved quickly without long-term effects (Tables 4 and 5).

4. Discussion

The findings of this randomized clinical trial demonstrate that TEAS significantly promotes gastrointestinal function recovery in adults aged 18–64 years undergoing elective laparoscopic non-gastrointestinal

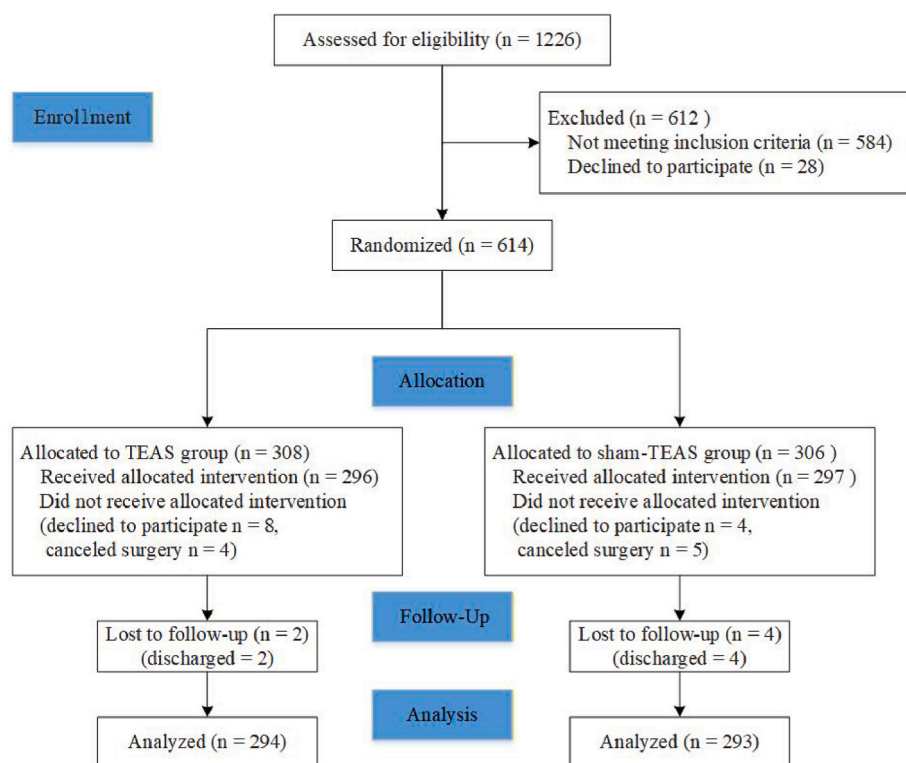


Fig. 1. CONSORT flow diagram. TEAS: transcutaneous electrical acupoint stimulation.

Table 1

Patients baseline characteristics.

Characteristics	TEAS (n = 294)	Sham-TEAS (n = 293)	P value
Age, median (IQR), y	51 (40–58)	50 (39–58)	^a 0.9112
Height, mean (SD), cm	166.2 (7.5)	165.8 (7.6)	^b 0.551
Weight, mean (SD), kg	69.6 (13.1)	70.5 (13.3)	^b 0.399
Sex, n (%)			
Male	122 (40.67)	115 (38.33)	^c 0.5588
Female	178 (59.33)	185 (61.67)	
ASA physical status classification, n (%)			
I	14 (2.4)	4 (1.4)	^c 0.264
II	250 (85.0)	257 (87.7)	
III	34 (11.6)	32 (10.9)	
BMI, n (%)			
Underweight (<18.5)	3 (1.0)	4 (1.4)	^c 0.564
Normal (18.5–23.9)	119 (40.5)	106 (36.2)	
Overweight (≥24)	172 (58.5)	183 (62.5)	
Comorbidities, n (%)			
Hypertension	30 (10.2)	26 (8.9)	^c 0.583
Diabetes mellitus			

P values are two-sided, and $p < 0.05$ was considered statistically significant. TEAS: transcutaneous electrical acupoint stimulation; IQR: interquartile range; SD: standard deviation.

ASA: American Society of Anesthesiologists; BMI: body mass index.

^a Based on the Mann–Whitney *U* test.

^b Based on the independent Student's *t*-test.

^c Based on the Chi-square test or Fisher's exact test.

surgery. Specifically, compared with sham-TEAS, the TEAS group exhibited significantly shorter times to first flatus and first defecation, as well as significantly lower incidence of 24 h PONV and abdominal distension.

These findings are consistent with those previous studies [14,25], showing that compared with sham-TEAS, intraoperative TEAS is associated with a shorter time to first flatus or a lower POI rate. Wang et al. [25] randomized 249 patients undergoing laparoscopic colorectal cancer surgery to either electroacupuncture (EA) or sham

Table 2

Surgery and anesthesia related records.

Characteristics	TEAS (n = 294)	Sham-TEAS (n = 293)	P value
Type of surgery, n (%)			
LC	141 (48.0)	152 (51.9)	^b 0.636
LCBDE + LC	23 (7.8)	25 (8.5)	
Laparoscopic gynecologic surgery	102 (34.7)	87 (29.7)	
Others	28 (9.5)	29 (9.9)	
Surgical time, median (IQR), min	61 (45–89)	58 (43–85)	^a 0.280
Anesthesia time, median (IQR), min	75.5 (57–102)	73 (56–100)	^a 0.314
Blood loss, median (IQR), ml	50.0 (20.0–50.0)	50.0 (20.0–50.0)	^a 0.664
Total intraoperative infusion, median (IQR), ml	1000.0 (1000.0–1000.0)	1000.0 (1000.0–1000.0)	^a 0.165
Intraoperative medication			
Ondansetron, n (%)	38 (12.9)	35 (11.9)	^b 0.719
Dexamethasone, n (%)	234 (79.6)	237 (80.9)	^b 0.693
Neostigmine, n (%)	168 (57.1)	178 (60.8)	^b 0.374
Atropine, n (%)	189 (64.3)	203 (69.3)	^b 0.199
Remifentanyl, median (IQR), µg	841.4 (593.85–1278.0)	795.2 (584.0–1228.2)	^a 0.284
Sufentanil, median (IQR), µg	21.0 (18.0–22.8)	21.0 (18.0–22.5)	^a 0.927
Patient controlled intravenous analgesia, n (%)	286 (97.3)	279 (95.2)	^b 0.190

P values are two-sided, and $p < 0.05$ was considered statistically significant. TEAS: transcutaneous electrical acupoint stimulation; LC: laparoscopic cholecystectomy; LCBDE: laparoscopic common bile duct exploration; IQR: interquartile range.

^a Based on the Mann–Whitney *U* test.

^b Based on the Chi-square test or Fisher's exact test.

electroacupuncture (SA) group. Their results showed that the EA group experienced shorter times to first defecation (76.4 h vs. 90.0 h), and first

Table 3
Primary outcome measures between TEAS and sham-TEAS groups.

Primary outcome	TEAS (n = 294)	Sham-TEAS (n = 293)	Median difference (95% CI)	^a P value
time to flatus, median (IQR), h	21.5 (19.0–23.5)	23.5 (21.0–26.0)	–2.5 (–3.0 to –1.8)	<0.001
time to flatus, median (IQR), h (PP analysis)	21.3 (19.0–23.5)	23.4 (21.0–26.0)	–2.0 (–2.9 to –1.5)	<0.001
time to flatus, median (IQR), h (ITT analysis)	21.3 (19.0–23.5)	23.4 (21.0–26.0)	–2.0 (–2.9 to –1.5)	<0.001

P values are two-sided, and $p < 0.05$ was considered statistically significant.
TEAS: transcutaneous electrical acupoint stimulation; CI: confidence interval;
IQR: interquartile range; PP: per protocol; ITT: intention to treat.

^a Based on the Mann–Whitney *U* test.

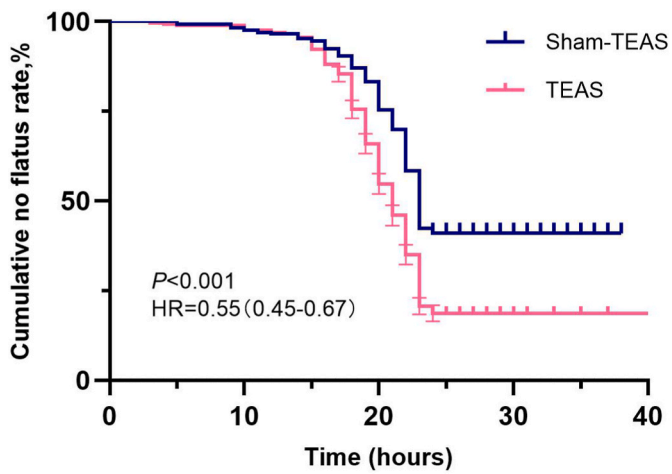


Fig. 2. Kaplan-Meier curves for cumulative no flatus rate.
TEAS
transcutaneous electrical acupoint stimulation; HR: hazard ratio.

flatus (44.3 h vs. 58.9 h), as well as earlier tolerance of a semiliquid diet (105.8 h vs. 116.5 h) and solid food (181.8 h vs. 190.3 h), compared with the SA group. In another multicenter randomized clinical trial of colorectal surgery, Gao et al. [19] reported that postoperative TEAS significantly reduced the rate of postoperative paralytic ileus after colorectal surgery by 9.7%. The intervention comprised four sessions: the first in the PACU upon reaching a Steward recovery score of 4, and the remaining three once daily between 9 and 11 a.m. on postoperative days 1–3 on the ward. However, definitions of postoperative ileus and gastrointestinal dysfunction remain unstandardized across studies investigating TEAS for postoperative gastrointestinal recovery.

POI is characterized by impaired gastrointestinal transit following abdominal surgery [7]. Despite significant efforts, a universally accepted definition of ileus has not been established. Variability in

Table 4
Secondary outcome measures in the TEAS and sham-TEAS groups.

Secondary outcomes	TEAS (n = 294)	Sham-TEAS (n = 293)	Effect size (95% CI)	P value
time to defecation, median (IQR), h	41.2 (38.5–45.0)	46.0 (42.0–52.0)	–4.78 (–3.8 to –6.0)	^a <0.001
24 h PONV, n (%)	44 (15)	67 (22.9)	0.59 (0.39–0.90)	^b 0.015
Incidence of abdominal distention, n (%)	62 (21.1)	87 (29.7)	0.63 (0.43–0.92)	^b 0.017
Hospital cost, median (IQR), Chinese yuan	22646.44 (20226.03–27808.88)	21957.81 (20099.88–25367.38)	–624.59 (–1309.47 to 24.33)	^a 0.059
Length of postoperative hospital stay, median (IQR)	3.0 (2–4)	3.0 (2–4)		^a 0.725

P values are two-sided, and $p < 0.05$ was considered statistically significant.
TEAS: transcutaneous electric acupoint stimulation; CI: confidence interval; IQR: interquartile range; PONV: postoperative nausea and vomiting; NRS: numeric rating scale.

^a Based on the Mann–Whitney *U* test.

^b Based on the Chi-square test or Fisher's exact test.

outcome reporting has been identified as a major source of heterogeneity in clinical studies. A systematic review [26] highlighted the clinical significance of time to first flatus as a reliable indicator of postoperative gastrointestinal recovery, noting its practicality compared with the assessment of bowel sounds, which clinicians often find challenging because of time constraints. Consequently, the time to first flatus and defecation remain key metrics for assessing postoperative gastrointestinal function and are commonly employed as endpoints in clinical trials [13,14,25,27]. We selected time to first flatus as the primary endpoint, which is consistent with established practice.

POGD is associated with a complex pathophysiological mechanism involving inflammation, fluid administration, medication effects, and ischemia-reperfusion injury [26,28,29]. TEAS is widely applied in clinical settings as an economical, non-invasive, and straightforward approach for managing gastrointestinal disorders, pain, stress urinary incontinence, and conditions related to reproductive medicine [17,20,30–33]. However, the underlying mechanisms of TEAS remain largely unclear. Factors such as the number of sessions, selection of acupoints, optimal timing, and environmental conditions influence the efficacy of electroacupuncture. In this study, the selected acupoints included Zusanli (ST36), Neiguan (PC6), and Hegu (LI4). The Zusanli (ST36) acupoint, located on the Foot Yang Ming stomach meridian, has an important effect on modulating the stomach and spleen while enhancing gastrointestinal motility. In modern mechanistic studies, TEAS at ST36 enhances gastrointestinal recovery by activating the vagus nerve and inhibiting inflammation, as evidenced by increased plasma levels of acetylcholine and IL-10 alongside decreased levels of IL-6 and iNOS [19]. Electroacupuncture at ST36 has been shown to relieve chronic constipation by increasing vagal activity and decreasing sympathetic activity [34]. Additionally, Liu et al. have reported that electroacupuncture at ST36 could relieve gastroesophageal reflux disease by modulating autonomic function [35]. The Neiguan (PC6) acupoint, a Luo-connecting point on the Pericardium Meridian of Hand Jueyin, can effectively alleviate gastrointestinal symptoms such as nausea and vomiting, likely through modulation of the vagus nerve [36]. Similarly, Hegu (LI4), a Yuan-source point on the Large Intestinal Meridian of Hand-Yangming [37], activates the vagal anti-inflammatory pathway,

Table 5
Postoperative complications and adverse events.

Adverse event	TEAS (n = 294)	Sham-TEAS (n = 293)	^a P value
Overall, n (%)	8 (2.7)	7 (2.4)	0.799
Severe adverse events, n (%)	0	0	
Subcutaneous hematoma, n (%)	0	0	0.995
Rubefaction, n (%)	6 (2.0)	6 (2.0)	0.565
Hydroderma, n (%)	2 (0.7)	1 (0.3)	

P values are two-sided, and $p < 0.05$ was considered statistically significant.
TEAS: transcutaneous electrical acupoint stimulation.

^a Based on the Chi-square test or Fisher's exact test.

thereby downregulating cytokine storms and exerting anti-inflammatory effects [38].

This study had several strengths, including large sample size and the use of a sham stimulus, which minimized subjective bias in the control group and ensured credible findings. However, there are some limitations. First, caution should be exercised when generalizing our findings to other populations or settings because of its single-center nature. Second, despite strict blinding protocols, the study did not evaluate whether the blinding was effective by asking patients to guess their assigned stimulation. Third, the use of fixed block randomization introduced potential bias due to treatment allocation predictability. Fourth, the precise impact of TEAS on gastrointestinal activity remains unclear, necessitating further studies to elucidate its mechanisms.

5. Conclusion

In young and middle-aged patients undergoing laparoscopic non-gastrointestinal surgery, TEAS significantly reduces time to first flatus and defecation, as well as the incidence of PONV and abdominal distension, without affecting hospital stay duration or costs. These findings suggest that TEAS may be a useful adjunctive therapy in this setting. Further research is needed to explore the mechanisms underlying such interventions and to validate these results across diverse populations and clinical settings.

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CRediT authorship contribution statement

Xianhong Jiang: Data curation, Methodology, Writing – original draft. **Yuan Zhang:** Conceptualization, Methodology. **Jiaqi Tan:** Formal analysis, Investigation. **Ping Zhuang:** Methodology. **Aili Fang:** Methodology. **Xiaomin Yue:** Data curation. **Jun Pang:** Data curation. **Ying Kong:** Data curation. **Ke Zhang:** Investigation. **Yanfang Zhang:** Investigation, Methodology. **Jianbo Yu:** Conceptualization, Methodology, Supervision.

Conflicts of interest

The authors have no competing interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ctcp.2026.102076>.

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